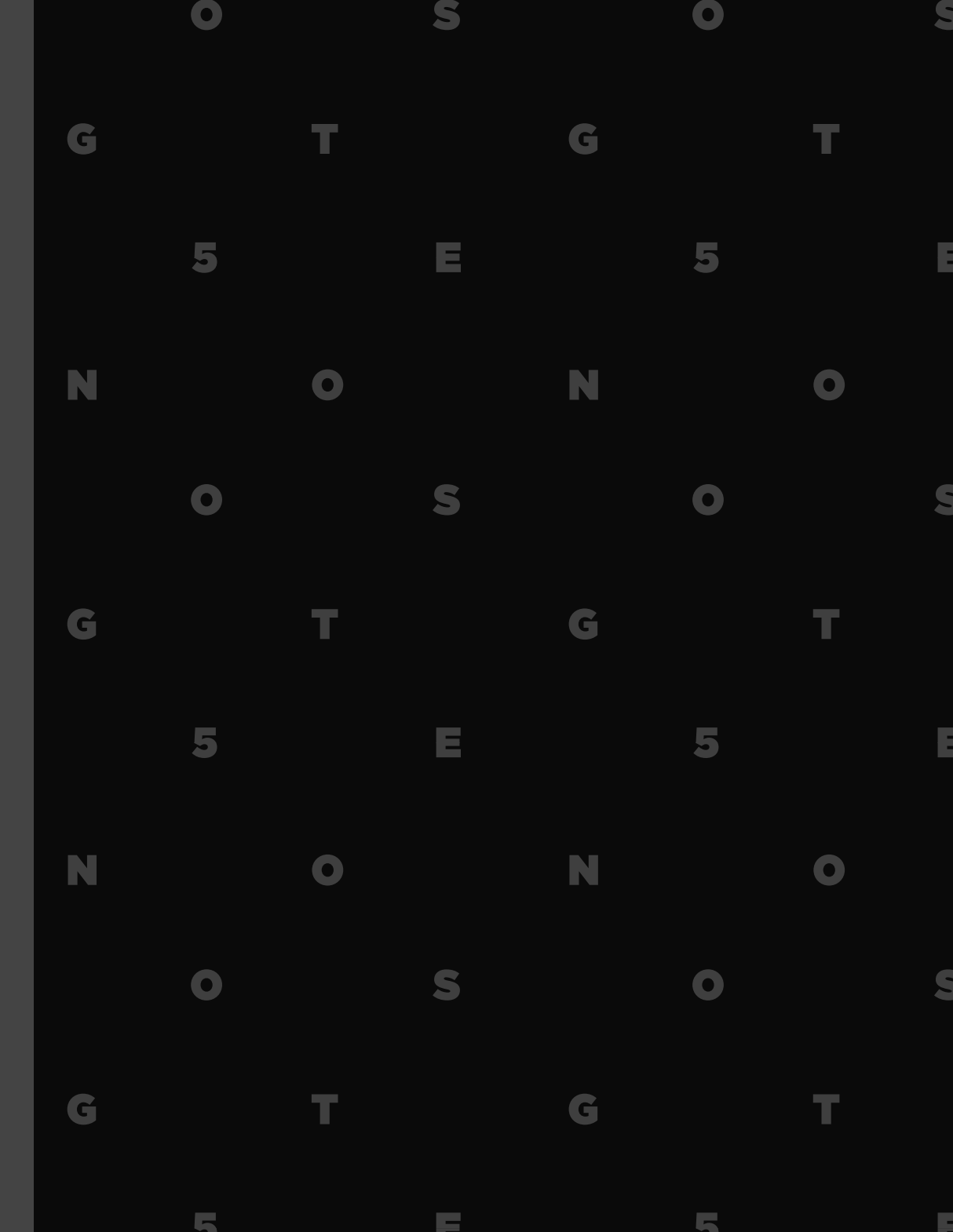
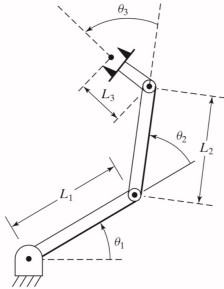




RMC Homework - 3

Kandiraju Venkata Sai Advaita

3.1)



i	a_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3

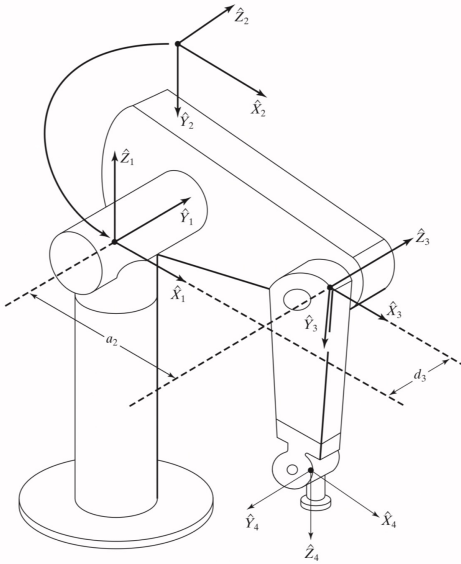
$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & L_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & L_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_3T = {}^0_1T \cdot {}^1_2T \cdot {}^2_3T \Rightarrow \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1 + L_2 c_{12} \\ s_{123} & c_{123} & 0 & L_1 s_1 + L_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2)



We can perform a 180° tilt on the prismatic arm to represent d_3 in the DH table.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	θ_2
3	90	0	d_3	180°
4	0	a_3	d_4	θ_4
5	90	0	0	θ_5
6	-90	0	0	θ_6

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3T_4 = \begin{bmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_6 & c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

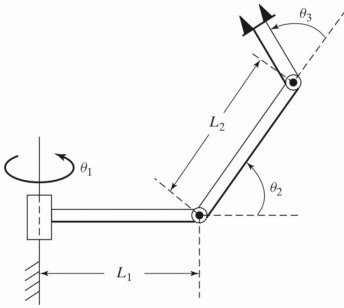
$${}^0_6T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T.$$

$$\Rightarrow \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where,

$$\begin{aligned} r_{11} &= -c_1c_2c_4c_5c_6 + c_1c_2s_4s_6 + s_1s_4c_5c_6 + s_1c_4s_6 + c_1s_2s_5s_6 \\ r_{12} &= c_1c_2c_4c_5s_6 + c_1c_2s_4c_6 - s_1s_4c_5c_6 + s_1c_4c_6 - s_1s_2s_5s_6 \\ r_{13} &= c_1c_2c_4s_5 - s_1s_4s_5 + c_1s_2c_5 \\ r_{21} &= -s_1c_2c_4c_5c_6 + s_1c_2s_4s_6 - c_1s_4c_5c_6 - c_1c_4s_6 + s_1s_2s_5c_6 \\ r_{22} &= s_1c_2c_4c_5c_6 + s_1c_2s_4s_6 + c_1s_4c_5c_6 - c_1c_4c_6 - s_1s_2s_5s_6 \\ r_{23} &= s_1c_2c_4s_5 + c_1c_4s_5 + s_1s_2c_5 \\ r_{31} &= s_2c_4c_5c_6 - s_2s_4s_6 + c_2s_5c_6 \\ r_{32} &= -s_2c_4c_5c_6 - s_2s_4c_6 - c_2s_5s_6 \\ r_{33} &= -s_2c_4c_5 + c_2c_5 \\ p_x &= (d_3 + d_4)c_1s_2 - a_3c_1c_2 - d_2s_1 \\ p_y &= (d_3 + d_4)s_1s_2 - a_3s_1c_2 + d_2c_1 \\ p_z &= (d_3 + d_4)c_2 + a_3s_2 \end{aligned}$$

3.3)



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	90°	L_1	0	θ_2
3	0	L_2	0	θ_3

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

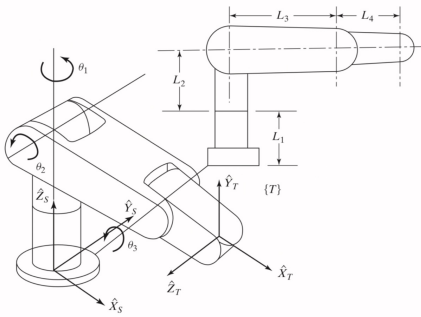
$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & L_1 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & L_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_3T = {}^0_1T {}^1_2T {}^2_3T \rightarrow {}^B_WT$$

$${}^B_WT \Rightarrow \begin{bmatrix} c_1 s_{23} & -c_1 s_{23} & s_1 & L_1 c_1 + L_2 c_1 c_2 \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & L_1 s_1 + L_2 s_1 c_2 \\ s_{23} & c_{23} & 0 & L_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3.4)



i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	$L_1 + L_2$	θ_1
2	90°	0	0	θ_2
3	0	L_3	0	θ_3
4	0	L_4	0	0

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & L_1 + L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & L_3 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} 1 & 0 & 0 & L_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_4T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T$$

$${}^2_4T = \begin{bmatrix} c_3 & -s_3 & 0 & c_3 L_4 + L_3 \\ s_3 & c_3 & 0 & L_4 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_2T = \begin{bmatrix} c_1 c_2 & -s_1 c_2 & s_1 & 0 \\ c_2 s_1 & -s_1 s_2 & -c_1 & 0 \\ s_2 & c_2 & 0 & L_1 + L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4.2) from 3.3,

$${}^0_3T = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1 (c_2 L_2 + L_1) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1 (c_2 L_2 + L_1) \\ s_{23} & c_{23} & 0 & s_2 L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$s_1 = r_{11} \quad , \quad -c_1 = r_{23}$$

$$\theta_1 = \tan^{-1} [r_{13} / (-r_{23})]$$

if $r_{12} = r_{23} = 0$, the goal can't be reached.

$$p_x = c_1 (c_2 L_2 + L_1)$$

$$p_y = s_1 (c_2 L_2 + L_1)$$

$$\text{then, } c_2 = (1/L_2) (p_x / c_1 - L_1) \quad \text{if } c_1 \neq 0$$

$$\text{or } c_2 = (1/L_2) (p_x / s_1 - L_1) \quad \text{if } s_1 = 0$$

$$\text{Also, } p_y = s_2 L_2$$

$$\theta_2 = \tan^{-1} \left(\frac{p_y / L_2}{c_2} \right)$$

$$\text{then, } s_{23} = r_{31} \quad \text{and} \quad c_{23} = r_{32}$$

$$\text{so, } t_{23} = r_{31} / r_{32}$$

$$\tan(\theta_2 + \theta_3) = r_{31} / r_{32}$$

$$\theta_3 = \tan^{-1} (r_{31} / r_{32}) - \theta_2$$

if $r_{31} = r_{32} = 0$, the goal can't be reached.